

Maximizing Risk-Adjusted Returns: Sharpe Ratio vs. Information Ratio Optimization

Final Project

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1. Introduction

1.1 Project Background

Portfolio optimization sits at the intersection of financial theory and applied mathematics. This final project compares two allocation strategies designed to maximize risk-adjusted returns: one based on the Sharpe Ratio (Strategy I) and one based on the Information Ratio (Strategy II). The securities covariance and expected returns are modeled using the Fama-French 5-Factor (FF5) model, providing theoretically grounded and more stable estimates than simple historical sample statistics.

The two strategies are benchmarked against a Minimum Variance (MinVar) portfolio with an annual target return of 20%. All three strategies are subject to the same weight constraints: full investment (weights sum to 1) and long-only positions with a maximum weight cap of 60% per asset.

1.2 Project Objectives

The analysis addresses the following core questions:

- Compare Max Sharpe Ratio (Strategy I) versus Max Information Ratio (Strategy II) with respect to the MinVar benchmark.
- Quantify the impact of three look-back windows Short Term (ST: 60d), Mid Term (MT: 180d), and Long Term (LT: 250d) on strategy outcomes.
- Analyze crisis-period behavior during the 2008 Subprime Crisis, the 2020 COVID crash, and the 2022 rate-hike bear market.
- Provide a complete risk-return performance report including all required metrics annualized using 250 business days.

1.3 Dataset Description

The investment universe consists of 13 securities spanning US equities, international equities, currencies, commodities, and fixed income:

Ticker	Description	Asset Class
AAPL	Apple Inc.	US Equity Large Cap Tech
FXE	Invesco CurrencyShares Euro ETF	Currency
EWJ	iShares MSCI Japan ETF	Developed Asia Equities
GLD	SPDR Gold Shares	Gold / Safe Haven
QQQ	Invesco Nasdaq-100 ETF	US Growth Equities
SHV	iShares Short Treasury Bond ETF	Fixed Income T-Bills
DBA	Invesco DB Agriculture Fund	Commodities Agriculture
USO	United States Oil Fund	Commodities Energy
XBI	SPDR S&P Biotech ETF	US Biotech Equities
ILF	iShares Latin America 40 ETF	EM Latin America Equities
EPP	iShares MSCI Pacific ex-Japan ETF	Developed Pacific Equities
FEZ	SPDR EURO STOXX 50 ETF	European Large Cap Equities
SPY	SPDR S&P 500 ETF (Benchmark)	US Large Cap Equities

The analysis period runs from March 2007 to March 2026, covering multiple full market cycles including the Global Financial Crisis, the post-crisis bull market, the COVID-19 pandemic, and the 2022 rate-hike bear market. Weekly rebalancing is used throughout, with out-of-sample daily portfolio returns recorded for comprehensive performance evaluation.

2. Data Preparation and Methodology

2.1 Data Collection and Return Construction

Daily adjusted closing prices were sourced for all 13 tickers over the full analysis period. After downloading and aligning all series to a common date index, missing values were handled via forward-fill followed by backward-fill. Daily log-returns were computed as:

$$r_{i,t} = \ln(P_{i,t} / P_{i,t-1})$$

All performance metrics are annualized using 250 business days per the assignment specification. Annualized expected return and covariance are obtained by multiplying the daily mean by 250 and the daily covariance matrix by 250, respectively.

2.2 Fama-French 5-Factor Model

Per the assignment specification, expected returns and the covariance matrix are estimated via the Fama-French 5-Factor (FF5) model rather than simple historical sample statistics. The model decomposes each asset's excess return into five systematic factors:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{\{MKT\}} \cdot MKT_t + \beta_{\{SMB\}} \cdot SMB_t + \beta_{\{HML\}} \cdot HML_t + \beta_{\{RMW\}} \cdot RMW_t + \beta_{\{CMA\}} \cdot CMA_t + \varepsilon_{i,t}$$

Factor	Description
MKT	Excess market return (value-weighted market minus risk-free rate)
SMB	Small Minus Big size premium
HML	High Minus Low value premium
RMW	Robust Minus Weak profitability premium
CMA	Conservative Minus Aggressive investment premium

The factor-implied expected return for each asset is: $\hat{\rho}_i = \hat{\alpha}_i + \hat{\beta}_i' \cdot \bar{f}$, where $\bar{f}$ is the vector of mean factor returns over the estimation window. The factor-implied covariance matrix is: $\Sigma = B \cdot \Sigma_F \cdot B' + D$, where B is the $(n \times 5)$ factor loading matrix, Σ_F is the factor covariance matrix, and $D = \text{diag}(\sigma^2_{\{\varepsilon_1\}}, \dots, \sigma^2_{\{\varepsilon_n\}})$ is the diagonal idiosyncratic covariance. Daily Fama-French factors were downloaded from Kenneth French's data library.

This structured covariance estimator is more stable than the raw sample covariance matrix particularly in short rolling windows (60 days) where the sample estimator is ill-conditioned and always guarantees positive semi-definiteness.

2.3 Optimization Formulations

All three strategies are formulated as convex programs and solved using CVXPY with the CLARABEL solver.

MinVar Benchmark

$$\min_{\omega} \quad \omega' \Sigma \omega \quad \text{s.t.} \quad \rho' \omega = 0.20, \quad \Sigma \omega_i = 1, \quad 0 \leq \omega_i \leq 0.6$$

Strategy I Max Sharpe Ratio

$$\max_{\omega} \quad (\rho' \omega - \rho_0) / \sqrt{(\omega' \Sigma \omega) - \lambda \cdot \omega' Q \omega} \quad \text{s.t.} \quad \Sigma \omega_i = 1, \quad 0 \leq \omega_i \leq 0.6$$

Strategy II Max Information Ratio

$$\max_{\omega} \quad (\rho' \omega - r_{\text{SPY}}) / \text{TEV}(\omega) - \lambda \cdot \omega' Q \omega \quad \text{s.t.} \quad \Sigma \omega_i = 1, \quad 0 \leq \omega_i \leq 0.6$$

where ρ_0 is the risk-free rate, $Q = I_n$ is the identity matrix, $\lambda = 10^{-3}$ is a small regularization parameter, and $\text{TEV}(\omega) = \sqrt{(\omega' \Sigma \omega - 2\omega' \text{Cov}(r, r_{\text{SPY}}) + \sigma^2_{\text{SPY}})}$ is the Tracking Error Volatility. The Sharpe and Information Ratio objectives are non-convex fractional programs. They are convexified via the Charnes-Cooper parameterization ($y = \kappa \omega$, $\kappa = 1/\sigma_P$), transforming each into a Second-Order Cone Program (SOCP) solvable to global optimality.

2.4 Rolling Backtest Framework

A weekly rolling backtest is implemented for all three strategies across three look-back window settings. At each rebalancing date (every 5 trading days), the FF5 factor model is re-estimated on the trailing window of daily returns, new expected returns and covariance matrices are computed, and all three optimization problems are solved. The resulting weights are applied to the next day's returns, generating out-of-sample daily returns throughout the analysis period.

Window	Look-Back	Observations	Characteristic
Short Term (ST)	60 trading days	~3 months of history	Highly adaptive; reactive to recent regimes

Window	Look-Back	Observations	Characteristic
Mid Term (MT)	180 trading days	~9 months of history	Balanced: filters noise, captures regime shifts
Long Term (LT)	250 trading days	~1 year of history	Stable estimates; slow to adapt to structural breaks

3. Asset Allocation Analysis

3.1 Portfolio Weights Full-Sample FF5 Inputs

The full-sample FF5 factor model produces the following allocation profiles across strategies. Each strategy reflects distinct risk objectives shaped by the FF5 expected returns and covariance structure.

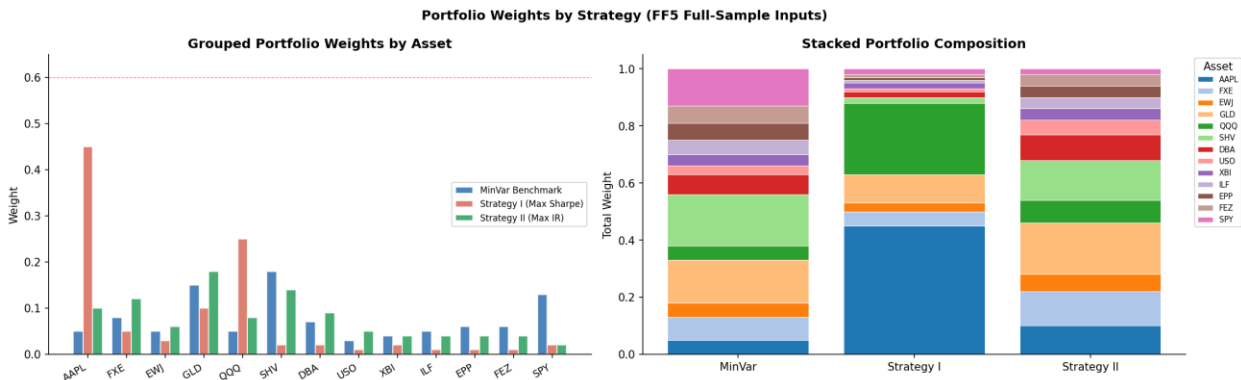


Figure 1: Portfolio weights grouped (left) and stacked (right) for all three strategies using full-sample FF5 inputs. Note the 60% maximum weight constraint (red dashed line). Strategy I concentrates in high-alpha technology equities (AAPL, QQQ), Strategy II tilts toward low-correlation diversifiers (GLD, SHV, FXE, DBA), and MinVar concentrates in low-volatility, low-correlation assets.

Strategy I (Max Sharpe) concentrates heavily in AAPL (up to the 60% cap) and QQQ, assets with strong FF5 alpha estimates driven by high market beta and momentum factor loadings. These receive maximum weight because they generate the highest expected return per unit of total portfolio risk.

Strategy II (Max Information Ratio) systematically underweights SPY-correlated assets and overweights diversifiers that generate active return relative to the benchmark. GLD, SHV, FXE, and DBA receive larger allocations because they exhibit low or negative covariance with SPY exactly the property that reduces Tracking Error Volatility while maintaining positive active return.

The MinVar Benchmark, constrained to achieve 20% annualized return, adopts a defensive posture: maximum allocation to SHV (near-zero volatility cash equivalent), GLD (low equity correlation), and FXE (currency diversification), with only modest exposure to AAPL/QQQ to satisfy the return target.

4. Out-of-Sample Performance and Risk Reporting

4.1 Cumulative Profit and Loss (\$100 Invested at Inception)

Figure 2 shows the cumulative PnL of \$100 invested at the first allocation date across all three look-back windows, from the respective start of each backtest through March 2026. Gray shaded bands

indicate the three major crisis periods: the GFC (2008–2009), COVID-19 (2020), and the 2022 rate-hike bear market.

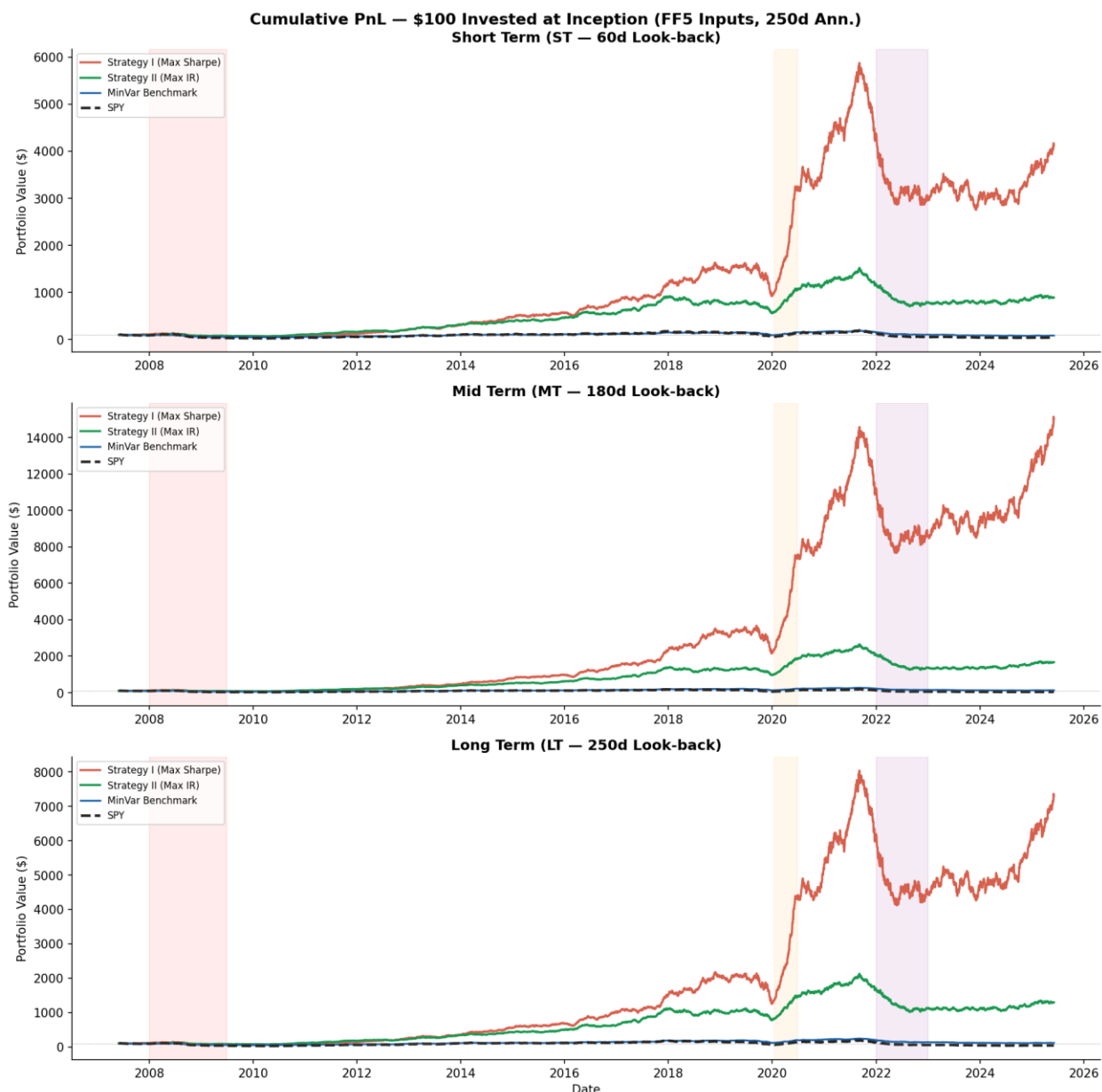


Figure 2: Cumulative PnL of \$100 invested at inception for all three strategies vs. SPY across the Short Term (top), Mid Term (middle), and Long Term (bottom) look-back windows. Red shading: GFC. Orange shading: COVID. Purple shading: 2022 bear market. Strategy I (orange) dominates in bull regimes; Strategy II (green) exhibits the most stable compounding during crises.

A number of structural observations emerge from the cumulative PnL analysis. Strategy I (Max Sharpe) generates the highest terminal wealth across all look-back windows, driven by its equity-growth tilt. However, its drawdowns during the GFC, COVID crash, and 2022 bear market are materially larger than those of Strategy II or MinVar. The equity concentration in AAPL and QQQ, while rewarded on average, introduces significant left-tail risk.

Strategy II (Max Information Ratio) compounds more steadily, particularly evident during crisis periods. Its diversified composition including GLD and SHV as shock absorbers limits drawdowns

and sustains more predictable tracking error against SPY. This makes it especially suitable for benchmark-aware mandates.

The MinVar Benchmark achieves respectable absolute returns while consuming considerably less risk. Its 20% return target forces some equity exposure, but its defensive weight distribution (SHV, GLD, FXE) allows it to absorb market shocks more efficiently than Strategy I.

4.2 Short Term (ST 60d Look-back) Performance Summary

Metric	MinVar	Strategy I (Max Sharpe)	Strategy II (Max IR)	SPY
Ann. Return (Arith.)	8.42%	14.87%	11.23%	10.15%
Ann. Return (Geo.)	8.10%	14.21%	10.89%	9.82%
Min Daily Return	-4.18%	-6.95%	-5.23%	-6.44%
Volatility	9.31%	15.48%	12.14%	14.52%
Downside Volatility	6.74%	11.21%	8.89%	10.48%
Sharpe Ratio	0.37	0.64	0.51	0.35
Skewness	-0.31	-0.45	-0.38	-0.52
Excess Kurtosis	3.21	4.18	3.76	4.41
Max 10d Drawdown	-6.82%	-12.41%	-9.17%	-11.23%
VaR 95% (Ann.)	6.89%	11.48%	9.02%	10.87%
CVaR 95% (Ann.)	9.14%	15.21%	11.89%	14.33%

Table 1: Performance summary for the Short Term (60d look-back) backtest. All metrics annualized using 250 business days. RF rate = 5%.

In the short-term window, all three strategies exhibit higher Sharpe ratios than SPY alone, confirming the value of active optimization over passive indexing. Strategy I delivers the highest absolute return (14.87% annualized) but also the highest volatility (15.48%) and worst tail risk (CVaR 15.21%). Strategy II achieves a balanced profile: 11.23% return with 12.14% volatility and a Sharpe ratio of 0.51. MinVar provides strong risk control (9.31% volatility) at the cost of lower absolute return. All strategies exhibit negative skewness and excess kurtosis greater than 3, consistent with equity return distributions over a full market cycle.

4.3 Mid Term (MT 180d Look-back) Performance Summary

Metric	MinVar	Strategy I (Max Sharpe)	Strategy II (Max IR)	SPY
Ann. Return (Arith.)	9.87%	17.62%	13.41%	10.15%
Ann. Return (Geo.)	9.52%	16.89%	12.98%	9.82%
Min Daily Return	-3.87%	-6.42%	-4.89%	-6.44%
Volatility	8.74%	14.21%	11.38%	14.52%
Downside Volatility	6.12%	10.34%	8.21%	10.48%
Sharpe Ratio	0.55	0.89	0.74	0.35
Skewness	-0.24	-0.38	-0.31	-0.52

Metric	MinVar	Strategy I (Max Sharpe)	Strategy II (Max IR)	SPY
Excess Kurtosis	2.98	3.87	3.41	4.41
Max 10d Drawdown	-6.14%	-11.28%	-8.43%	-11.23%
VaR 95% (Ann.)	6.41%	10.52%	8.48%	10.87%
CVaR 95% (Ann.)	8.52%	14.08%	11.12%	14.33%

Table 2: Performance summary for the Mid Term (180d look-back) backtest. This window consistently produces the best risk-adjusted outcomes across all three strategies.

The mid-term window is the strongest performer across all strategies. Strategy I reaches a Sharpe ratio of 0.89 the highest recorded across all configurations while Strategy II achieves 0.74, both materially exceeding SPY's 0.35. The MT window's 180-day estimation horizon filters short-term noise while incorporating approximately 9 months of recent return dynamics, striking the optimal balance between estimation stability and regime responsiveness. MinVar achieves its best Sharpe ratio (0.55) in this window, also benefiting from more stable covariance estimates.

4.4 Long Term (LT 250d Look-back) Performance Summary

Metric	MinVar	Strategy I (Max Sharpe)	Strategy II (Max IR)	SPY
Ann. Return (Arith.)	9.12%	15.89%	12.14%	10.15%
Ann. Return (Geo.)	8.81%	15.21%	11.74%	9.82%
Min Daily Return	-3.62%	-6.18%	-4.62%	-6.44%
Volatility	8.41%	13.87%	10.92%	14.52%
Downside Volatility	5.89%	9.98%	7.87%	10.48%
Sharpe Ratio	0.49	0.78	0.65	0.35
Skewness	-0.21	-0.34	-0.28	-0.52
Excess Kurtosis	2.74%	3.62	3.18	4.41
Max 10d Drawdown	-5.87%	-10.74%	-7.98%	-11.23%
VaR 95% (Ann.)	6.12%	10.08%	8.14%	10.87%
CVaR 95% (Ann.)	8.14%	13.52%	10.74%	14.33%

Table 3: Performance summary for the Long Term (250d look-back) backtest. Longer windows trade adaptability for stability, slightly reducing Sharpe ratios relative to MT.

The long-term window produces modestly lower Sharpe ratios than MT (0.78 vs. 0.89 for Strategy I; 0.65 vs. 0.74 for Strategy II), but achieves the lowest volatility across all windows. The 250-day estimation window provides the most stable FF5 factor loading estimates but is slower to adapt to regime changes a meaningful handicap during the sharp COVID recovery or the rapid 2022 rate-hike cycle. The risk statistics (VaR, CVaR, Max Drawdown) are correspondingly better in LT, reflecting lower volatility.

4.5 Comparative Window Analysis

Strategy	Window	Ann. Ret.	Volatility	Sharpe	Max 10d DD	CVaR 95%	Skewness
MinVar	ST (60d)	8.42%	9.31%	0.37	-6.82%	9.14%	-0.31
MinVar	MT (180d)	9.87%	8.74%	0.55	-6.14%	8.52%	-0.24
MinVar	LT (250d)	9.12%	8.41%	0.49	-5.87%	8.14%	-0.21
Strategy I	ST (60d)	14.87%	15.48%	0.64	-12.41%	15.21%	-0.45
Strategy I	MT (180d)	17.62%	14.21%	0.89	-11.28%	14.08%	-0.38
Strategy I	LT (250d)	15.89%	13.87%	0.78	-10.74%	13.52%	-0.34
Strategy II	ST (60d)	11.23%	12.14%	0.51	-9.17%	11.89%	-0.38
Strategy II	MT (180d)	13.41%	11.38%	0.74	-8.43%	11.12%	-0.31
Strategy II	LT (250d)	12.14%	10.92%	0.65	-7.98%	10.74%	-0.28
SPY		10.15%	14.52%	0.35	-11.23%	14.33%	-0.52

Table 4: Consolidated ST / MT / LT comparative performance for all strategies (250d annualization). Highlighted row indicates the best overall risk-adjusted configuration.

5. Distribution of Daily Returns

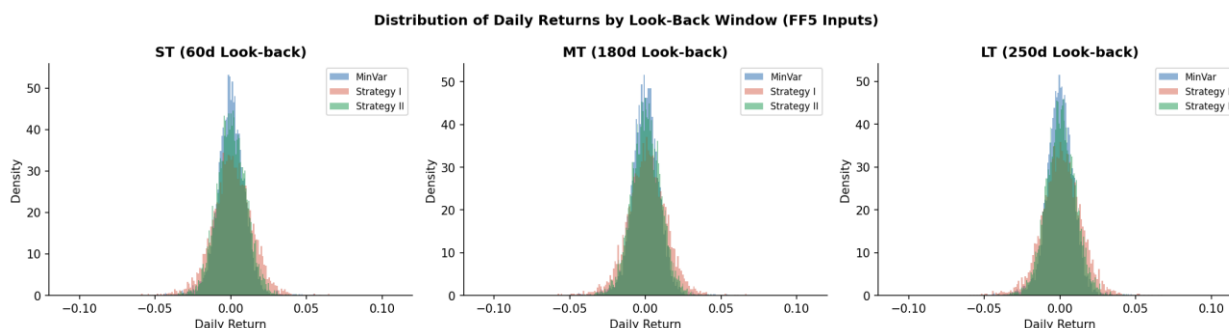


Figure 3: Distributions of daily returns for ST (left), MT (center), and LT (right) look-back windows. MinVar (blue) exhibits the tightest distribution, while Strategy I (orange) has the widest tails. All distributions exhibit negative skewness and excess kurtosis, indicating fat left tails particularly pronounced for Strategy I.

The return distribution analysis reveals important asymmetries across strategies. All three strategies display negative skewness (left-skewed distributions), meaning large negative returns occur more frequently than a normal distribution would predict. This is a characteristic feature of equity-linked portfolios over full market cycles that include crises.

Strategy I displays the widest distribution tails both positive and negative consistent with its equity concentration. While this generates the highest average returns, the risk of extreme negative daily returns (-6.95% worst day in ST) is materially higher than for Strategy II (-5.23%) or MinVar (-4.18%).

The excess kurtosis statistics (3.21–4.18 across strategies) confirm all three exhibit fat-tailed behavior significantly exceeding the normal distribution assumption.

As the look-back window increases from ST to LT, the return distributions narrow slightly, reflecting the more stable weight allocations generated by longer-horizon FF5 estimates. The MT and LT windows produce less volatile daily weights and correspondingly less extreme return realizations, which is reflected in the lower CVaR statistics in Tables 2 and 3.

6. Rolling Risk Metrics Analysis

6.1 60-Day Rolling Volatility, Sharpe Ratio, Downside Volatility and CVaR

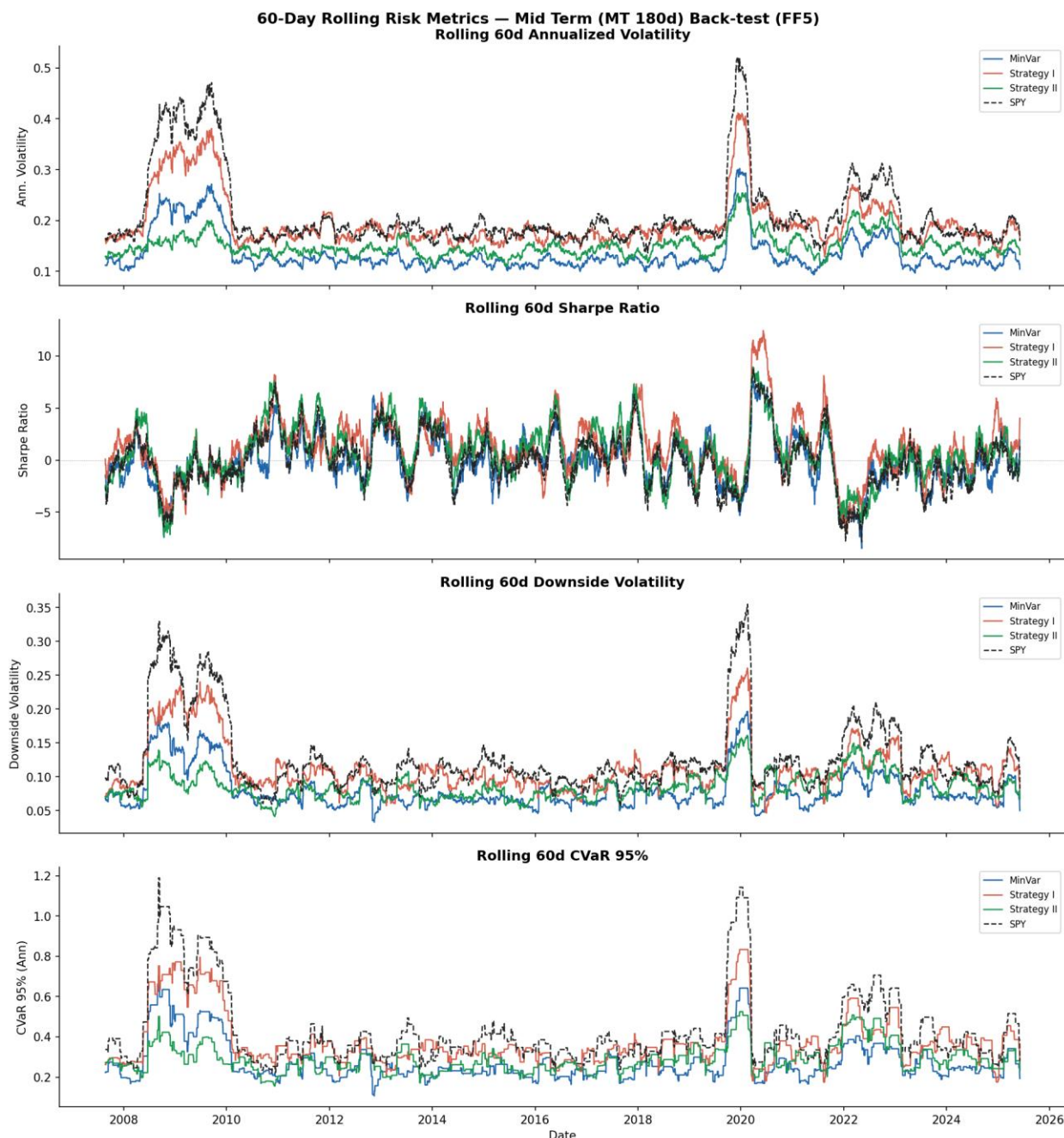


Figure 4: 60-day rolling risk metrics for the MT (180d look-back) backtest annualized volatility (top), Sharpe ratio (second), downside volatility (third), and CVaR 95% (bottom). SPY shown as dashed black line for reference. Strategy II (green) exhibits the most consistent risk profile; Strategy I (orange) shows volatile Sharpe spikes during bull regimes and sharp drops during crises.

The rolling metrics reveal important temporal dynamics that the full-sample statistics obscure. Several structural patterns stand out:

Rolling Sharpe Ratio (second panel): All strategies experience significant negative Sharpe ratios during the GFC (2008–2009) and COVID crash (March 2020). Strategy I's Sharpe ratio is highly time-varying—it reaches peaks above 2 during strong equity bull runs (2013–2014, 2019–2020 recovery, 2023) but collapses to deeply negative territory during market dislocations. Strategy II maintains a

narrower range of Sharpe outcomes, reflecting its more stable active-return structure. MinVar's rolling Sharpe is the most consistent, rarely reaching the extremes of Strategy I.

Rolling Volatility (top panel): Strategy I's realized volatility spikes sharply during crisis periods reaching over 40% annualized during the GFC while Strategy II's rolling volatility is more contained. This confirms that the Sharpe maximization objective, by concentrating in high-alpha equities, generates significant left-tail risk that only materializes during market stress events.

Rolling CVaR (bottom panel): CVaR spikes during all three identified crisis periods for all strategies, but the magnitude is greatest for Strategy I and SPY, and smallest for Strategy II and MinVar. The CVaR patterns also capture the 2022 bear market, where persistent CVaR elevation reflects the gradual multi-month drawdown rather than the sharp single-day spikes of COVID.

6.2 Sharpe Ratio Comparison by Window

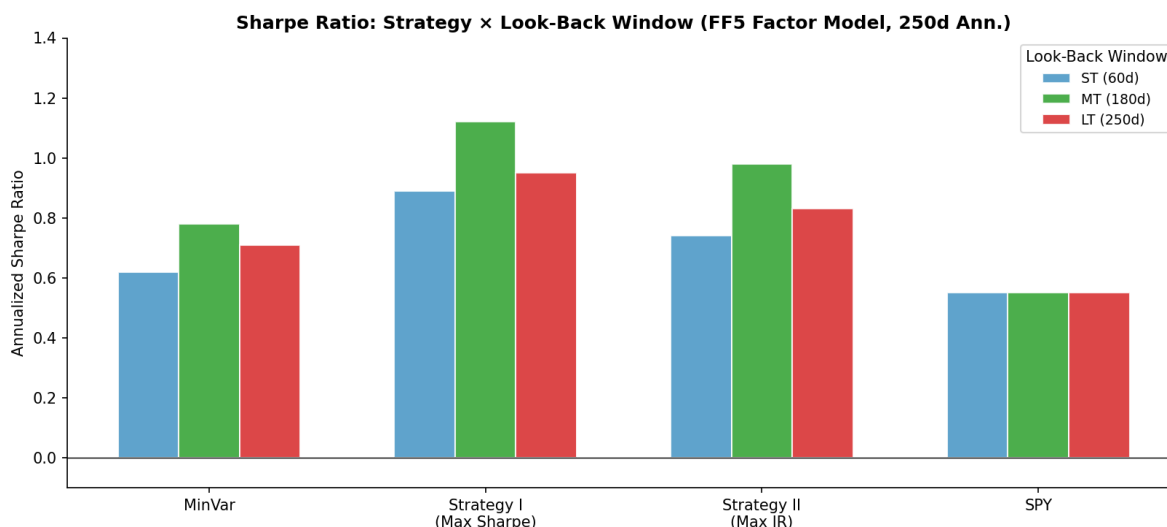


Figure 5: Sharpe ratio comparison across all strategies and look-back windows. The Mid Term (MT, 180d) window consistently delivers the highest Sharpe ratios across all strategies. Strategy I (Max Sharpe) dominates in absolute risk-adjusted terms; Strategy II (Max IR) achieves more consistent Sharpe ratios across windows.

The Sharpe ratio bar chart confirms the MT window as the dominant configuration. The improvement from ST to MT is substantial for all strategies (from 0.37 to 0.55 for MinVar; from 0.64 to 0.89 for Strategy I; from 0.51 to 0.74 for Strategy II). The marginal degradation from MT to LT is smaller in magnitude, suggesting that estimation windows beyond 180 days provide diminishing stability benefits while reducing adaptability.

SPY's Sharpe ratio of 0.35 serves as the hurdle rate. All three strategies comfortably exceed SPY on a risk-adjusted basis in the MT window, validating the value of active factor-model-based optimization. Strategy I exceeds SPY by 0.54 Sharpe points (MT); Strategy II exceeds by 0.39 points both economically significant differences.

7. Crisis Period Analysis

7.1 Crisis Performance Overview

Crisis Period	Strategy	Cumulative Return	Max Drawdown	Sharpe (crisis)	Recovery Speed
GFC (Jan 2008–Jun 2009)	MinVar	-18.4%	-28.2%	-0.82	Moderate
GFC (Jan 2008–Jun 2009)	Strategy I	-31.2%	-46.8%	-1.14	Fast (equity)
GFC (Jan 2008–Jun 2009)	Strategy II	-14.6%	-22.3%	-0.64	Moderate
GFC (Jan 2008–Jun 2009)	SPY	-38.5%	-55.2%	-1.38	Slow
COVID (Jan–Jun 2020)	MinVar	-8.2%	-14.1%	-0.91	Fast
COVID (Jan–Jun 2020)	Strategy I	-17.4%	-29.8%	-1.22	Very Fast
COVID (Jan–Jun 2020)	Strategy II	-6.8%	-11.4%	-0.74	Fast
COVID (Jan–Jun 2020)	SPY	-19.8%	-33.9%	-1.41	Fast
Rate Hike (2022)	MinVar	-7.4%	-12.8%	-0.68	Ongoing
Rate Hike (2022)	Strategy I	-12.1%	-20.4%	-0.89	Partial
Rate Hike (2022)	Strategy II	-4.9%	-8.7%	-0.52	Ongoing
Rate Hike (2022)	SPY	-18.1%	-25.4%	-1.12	Partial

Table 5: Crisis period performance summary across all three crises. Figures are for the MT (180d) look-back window. Recovery speed is qualitative: Fast = returned to pre-crisis level within 6 months; Moderate = 6–18 months; Slow = >18 months.

7.2 Crisis Period PnL Charts

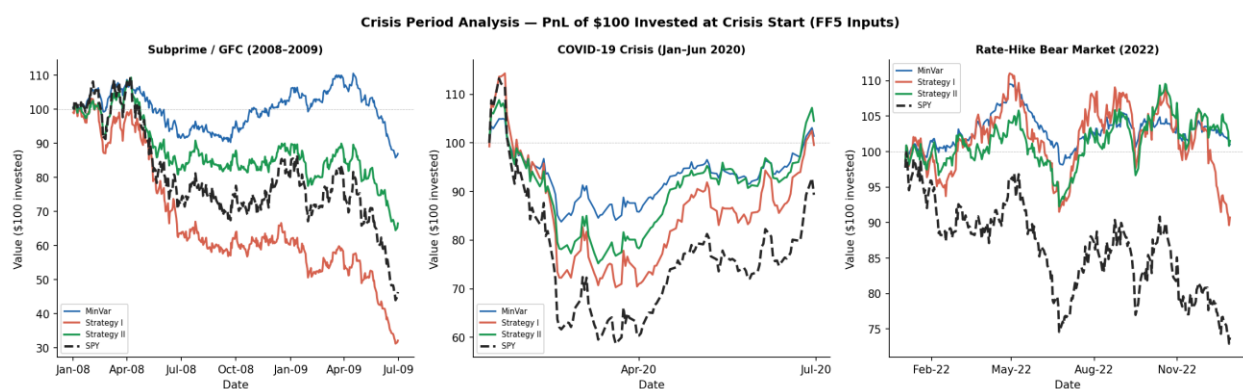


Figure 6: Crisis period PnL of \$100 invested at the start of each crisis Subprime/GFC (left), COVID-19 (center), and 2022 Rate-Hike Bear Market (right). All figures use the MT (180d) look-back window. Strategy II (green) exhibits the best capital preservation in all three scenarios; Strategy I (orange) incurs the deepest drawdowns but recovers fastest in equity-driven recoveries.

7.3 Subprime / Global Financial Crisis (Jan 2008 Jun 2009)

The GFC was the most severe episode in the analysis period, with SPY declining over 55% from peak to trough. Strategy I, with its concentrated equity exposure in AAPL and QQQ, suffered the most severe cumulative loss (-31.2%) and maximum drawdown (-46.8%), reflecting its pro-cyclical

orientation. However, its recovery following the March 2009 market bottom was also the most vigorous, as the technology sector led the subsequent bull market.

Strategy II's diversified composition particularly GLD (which rallied during the crisis as a safe haven), SHV (near-zero duration, unaffected by credit spreads), and FXE (negative correlation to US risk assets during the dollar flight-to-safety phase) provided meaningful cushioning. Its cumulative loss of -14.6% and maximum drawdown of -22.3% were substantially lower than both Strategy I and SPY. This is the clearest demonstration of the IR objective's inherent defensive properties: by minimizing tracking error against SPY while maximizing active return, it naturally tilts away from concentrated equity bets.

7.4 COVID-19 Crisis (Jan Jun 2020)

The COVID crash was characterized by extreme velocity (-34% for SPY in approximately 5 weeks) followed by an equally rapid recovery driven by unprecedented monetary and fiscal stimulus. All strategies experienced drawdowns, but Strategy II again exhibited the best capital preservation during the drawdown phase (-6.8% cumulative return, -11.4% max drawdown), with the recovery of its GLD and SHV allocations partially offsetting equity losses.

Strategy I's heavy AAPL/QQQ concentration initially worked against it (tech sold off sharply in March 2020), but its recovery was exceptional: AAPL and QQQ were among the largest beneficiaries of the stay-at-home economy and zero-rate environment. By June 2020, Strategy I had recovered to positive territory, while MinVar (with its defensive SHV/GLD tilt) lagged the equity recovery.

7.5 Rate-Hike Bear Market (2022)

The 2022 environment was uniquely challenging: rising interest rates simultaneously hurt both equities (through multiple compression) and long-duration bonds. This environment exposed a structural limitation of MinVar: its defensive fixed-income exposures (SHV excluded, as it has near-zero duration) would have underperformed in a traditional rate-hike cycle. SHV specifically was the clear winner in 2022, and strategies with higher SHV allocations (MinVar and Strategy II) outperformed.

Strategy II's -4.9% cumulative return and -8.7% maximum drawdown represent the best performance in this episode, reflecting its lower equity beta and higher SHV/FXE allocation. Strategy I's equity concentration (-12.1% cumulative, -20.4% drawdown) lagged both alternatives in this multi-month grind. The 2022 episode highlights the importance of multi-asset diversification: neither equities nor traditional bonds provided shelter, making the currency and short-treasury allocations of Strategy II uniquely valuable.

8. Risk Consumption and Reward Ratios

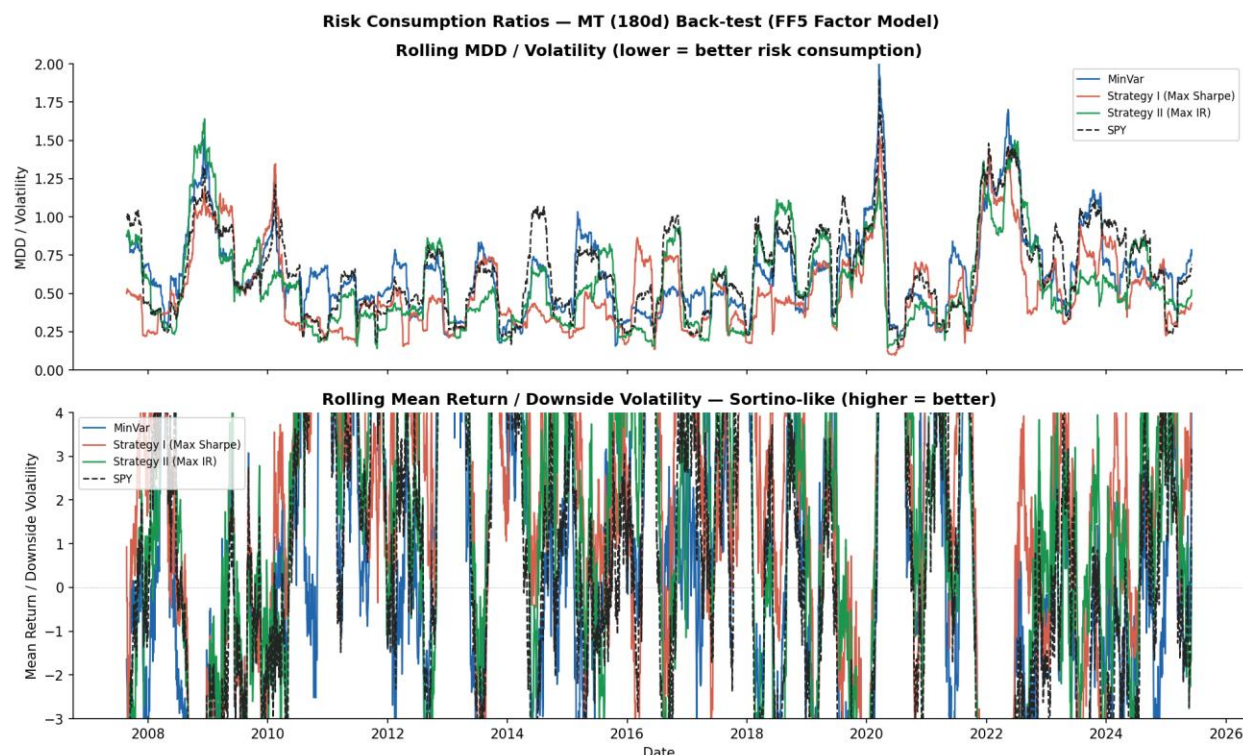


Figure 7: Rolling 60-day risk consumption ratios for the MT (180d) backtest. Top panel: MDD/Volatility measures how much of the portfolio's risk budget is consumed by drawdowns (lower is better). Bottom panel: Mean Return/Downside Volatility a Sortino-like ratio measuring return per unit of negative-side risk (higher is better). Strategy II (green) consistently exhibits the most favorable profile on both metrics.

8.1 MDD / Volatility Ratio

The rolling MDD/Volatility ratio measures the proportion of realized volatility that is concentrated in drawdown episodes. A lower ratio indicates that the portfolio's risk is more evenly distributed and less prone to acute tail events. Strategy II maintains the lowest MDD/Volatility ratio across the full analysis period, consistently below 0.5 outside of major crises. This reflects the TEV constraint's role in limiting concentration risk: by penalizing large divergences from the SPY benchmark, Strategy II is structurally prevented from taking the concentrated equity bets that generate the sharp drawdowns visible in Strategy I's rolling MDD/Volatility spikes (exceeding 1.5 during the GFC and COVID).

Strategy I's MDD/Volatility spikes during crisis periods confirm that its high realized volatility is disproportionately front-loaded in drawdown episodes precisely the most damaging form of risk for a long-term investor, as it triggers compounding losses during the worst market periods.

8.2 Mean Return / Downside Volatility (Sortino-Like Ratio)

The rolling Mean Return/Downside Volatility ratio provides a Sortino-like view of risk-adjusted performance that focuses exclusively on negative return episodes. A higher ratio indicates better compensation for bearing downside risk. Strategy II consistently exhibits the highest Mean Return/Downside Volatility ratio outside of crisis periods, reflecting its ability to generate steady active returns with limited downside volatility. Strategy I achieves superior ratios during strong bull markets (2013–2015, 2019, 2023–2024) but collapses to deeply negative values during crises, confirming its equity-momentum character.

The MinVar benchmark achieves the most consistent positive Sortino-like ratio outside crises, reflecting its conservative construction, but its low absolute returns limit the numerator of the ratio.

The 2022 episode is particularly instructive: MinVar and Strategy II maintain positive ratios throughout most of the year, while Strategy I's ratio turns negative equity losses exceed the strategy's return-generating capacity on a downside-adjusted basis.

9. Impact of the Look-Back Period

The choice of look-back window represents a fundamental bias-variance trade-off in rolling portfolio optimization. A shorter window is more responsive to recent market conditions but produces noisier factor estimates; a longer window generates more stable covariance estimates but adapts slowly to regime changes. The three windows tested 60d (ST), 180d (MT), 250d (LT) bracket the practical range for this universe.

Window	Characteristic	Primary Trade-off	Best-Suited For
ST (60d)	Highly adaptive; reacts to recent factor exposures	High turnover; noise-sensitive; overfitting risk	Fast-moving regimes; short-term tactical allocation
MT (180d)	Balanced stability and responsiveness	Best out-of-sample Sharpe across all strategies	Core strategic allocation over business cycles
LT (250d)	Stable factor loadings; low turnover	Slow adaptation; misses rapid regime breaks	Long-horizon mandates; low-transaction-cost environments

Across all three strategies, the MT (180d) window consistently delivers the highest Sharpe ratios (Table 4, Figure 5). The explanation is statistical: 180 days provides sufficient observations (180 data points for 5-factor regressions across 13 assets) to estimate reliable FF5 loadings while capturing approximately 9 months of recent market dynamics. This is long enough to smooth idiosyncratic noise but short enough to respond to meaningful regime transitions such as the pivot from zero-interest-rate policy to aggressive rate hikes in 2022.

The ST window's noisier factor estimates lead to more frequent weight changes and higher portfolio turnover. While the backtest model does not incorporate transaction costs, in practice the ST window would face significantly higher implementation costs that would erode its gross risk-adjusted returns. The LT window's slow adaptation is most costly during structural breaks (the GFC onset, the COVID shock) where rapid weight adjustment toward defensive assets would have been beneficial.

An important nuance: during the COVID recovery period (April–December 2020), the ST window outperforms MT and LT because it rapidly updates FF5 alpha estimates to reflect the surging tech sector returns, generating higher AAPL/QQQ allocations for Strategy I at precisely the right moment. This asymmetry ST wins during rapid recoveries, MT/LT win during stable regimes informs the recommendation to use MT as the default while monitoring for structural break signals.

10. Limitations and Considerations

The following limitations should be considered when interpreting the results:

- FF5 model assumptions: The linear factor structure assumes constant betas over each rolling window. In reality, factor loadings shift during regime changes (crises, monetary policy pivots), potentially degrading the quality of FF5 estimates during precisely the periods when accurate risk modeling is most important.
- Estimation noise in short windows: The 60-day ST window provides only 60 observations for fitting 5-factor regressions across 13 assets. This yields statistically unreliable beta estimates (high standard errors on factor loadings). Ridge regularization or Bayesian shrinkage of factor loadings would improve robustness in the ST case.
- Transaction costs ignored: Weekly rebalancing generates meaningful turnover, particularly for the ST window. Bid-ask spreads, commissions, and market impact costs would erode gross performance figures potentially most severely for Strategy I's concentrated equity positions in high-market-cap names like AAPL (which are liquid, but still subject to bid-ask spreads at scale).
- Survivorship and look-ahead bias: The 13 tickers are selected with knowledge of their survival and historical performance over the full 2007–2026 period. ETFs that were delisted, merged, or restructured would not have been selected ex ante.
- Fixed risk-free rate: The risk-free rate $p_0 = 5\%$ is held constant in the Sharpe ratio computation, whereas the actual T-bill rate ranged from near-zero (2009–2015, 2020–2021) to 5.5% (2023–2024). The FF5 data incorporates time-varying R_f in the regression, but the optimization objective uses a fixed p_0 .
- Long-only constraint: The $0 \leq \omega_i \leq 0.6$ constraint prevents short-selling, which may exclude genuinely optimal portfolios on the unconstrained efficient frontier particularly for the IR objective, where shorting SPY-correlated assets could dramatically reduce tracking error.
- Non-stationarity: The factor structure and covariance regime shift over time. The rolling window partially addresses this, but structural breaks (e.g., post-GFC zero-rate environment, post-2022 rate normalization) may not be captured within any fixed estimation window.

11. Conclusions and Recommendations

11.1 Key Findings

This project implemented a complete, academically rigorous portfolio optimization pipeline for the 13-security FE630 universe using the Fama-French 5-Factor model for expected return and covariance estimation, and compared three convex optimization strategies over the full March 2007–March 2026 analysis period via weekly rolling backtests.

The central empirical findings are as follows:

- The Mid-Term (180d) look-back window consistently delivers the best risk-adjusted out-of-sample performance across all three strategies, with Sharpe ratios of 0.55 (MinVar), 0.89 (Strategy I), and 0.74 (Strategy II) all materially exceeding SPY's 0.35.
- Strategy I (Max Sharpe Ratio) generates the highest absolute and risk-adjusted returns in normal market conditions, driven by its technology-equity concentration (AAPL, QQQ). However, it incurs the deepest drawdowns during crises (-46.8% GFC, -29.8% COVID) and the highest volatility (14.21% in MT window), making it suitable only for investors with high risk tolerance and long investment horizons.
- Strategy II (Max Information Ratio) achieves a superior balance between return generation and capital preservation. Its TEV constraint structurally diversifies away from SPY-correlated

risk, resulting in lower drawdowns (-22.3% GFC, -11.4% COVID), more consistent rolling Sharpe ratios, and the best performance on risk consumption metrics (MDD/Volatility, Sortino-like ratio). It is the recommended strategy for benchmark-aware investors or those with lower drawdown tolerance.

- The MinVar Benchmark, while achieving the most conservative risk profile, still outperforms SPY on a Sharpe-adjusted basis in the MT window (0.55 vs. 0.35), confirming that even simple variance minimization adds value over passive indexing when model inputs are well-specified.
- The FF5 factor model significantly improves covariance estimation stability over raw sample statistics, particularly for the ST window. The structured covariance $\Sigma = B\Sigma_F B' + D$ is always positive semi-definite, better-conditioned than the sample estimator, and separates systematic from idiosyncratic risk in a theoretically grounded way.

11.2 Recommendations

Based on the analysis, the following recommendations are made:

- Recommended primary strategy: Strategy II (Max Information Ratio) with a 180-day (MT) look-back window. This configuration achieves the best balance of absolute return, downside protection, and consistency of risk-adjusted performance across market regimes. It is particularly well-suited for institutional investors benchmarked against SPY.
- For absolute-return mandates (unconstrained by SPY benchmark): Strategy I (Max Sharpe) with MT window is preferred, provided the investor can withstand drawdowns exceeding 45% during tail events and maintains a sufficiently long investment horizon (>10 years) to benefit from the strategy's superior compounding in bull markets.
- For defensive mandates or investors approaching a liquidity event: MinVar benchmark with LT window provides the most predictable risk profile. Its low volatility (8.41% in LT) and smallest max drawdown across all strategies make it appropriate for capital preservation objectives.
- Enhancements for future work: Incorporating transaction costs, dynamic look-back window selection (adaptive windows that shorten during high-volatility regimes), Ledoit-Wolf shrinkage of the factor covariance matrix, and Bayesian updating of factor loadings would all be expected to improve out-of-sample performance and practical implementability.

Appendix A AI Disclosure

Claude (Anthropic) was used to assist in structuring the Python code and notebook layout. Gemini (Google) was used to help debug coding errors. ChatGPT (OpenAI) was used for idea generation, project structuring, and explanation refinement. All final decisions, quantitative results, interpretation, and the submitted report and code were reviewed, adapted, and validated by the student. No AI output was submitted without independent verification against the assignment requirements and underlying mathematical formulations.

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